

Coding & Solution

Date	15 March 2026
Team ID	
Project Name	OptiCrop
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/DeepakJaggupalli/Opticrop
Programming Language(s)	Python 3.10
Framework(s) Used	Flask (backend), scikit-learn (ML)
Key Features Implemented	Crop recommendation, real-time prediction API, glassmorphism UI, mini-game, error handling.
Pending / Incomplete Features	Live weather-API auto-fill; fertilizer recommendation (future work).
Setup / Run Instructions	cd '5. Project Development Phase'; pip install -r requirements.txt; python app.py -> http://127.0.0.1:5000/

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

Code follows PEP 8, is modular (app.py + model_pipeline.py), documented, and deployed on Vercel. Model test accuracy 97.27%.